BIO102 Help Session - 4

Midsem-2 Paper Discussion

3. Red–green colour blindness is a sex-linked recessive trait in humans. In a family, the father and mother have normal red–green colour vision. If their son is colour-blind and their daughter has normal colour vision, give the genotypes of all members of the family. (2 marks)

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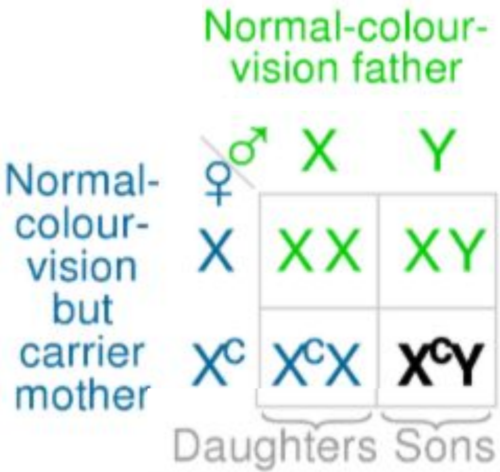
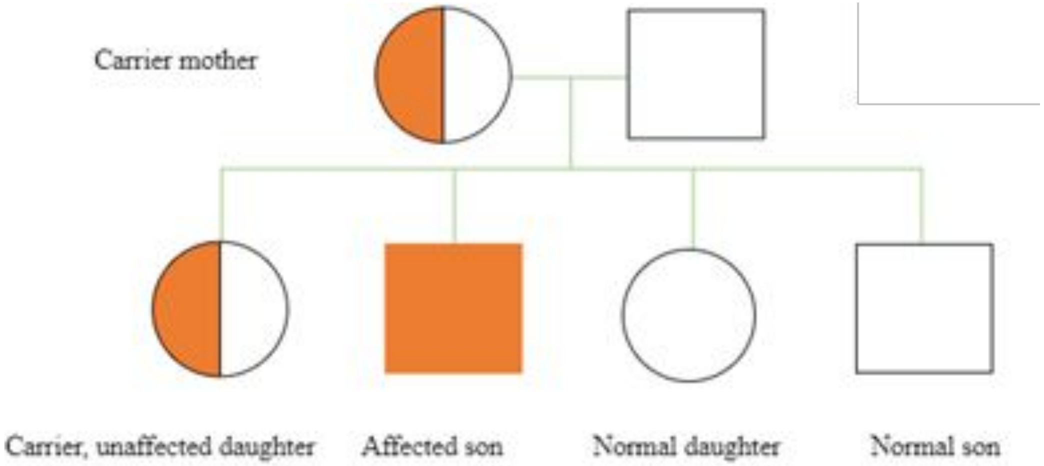
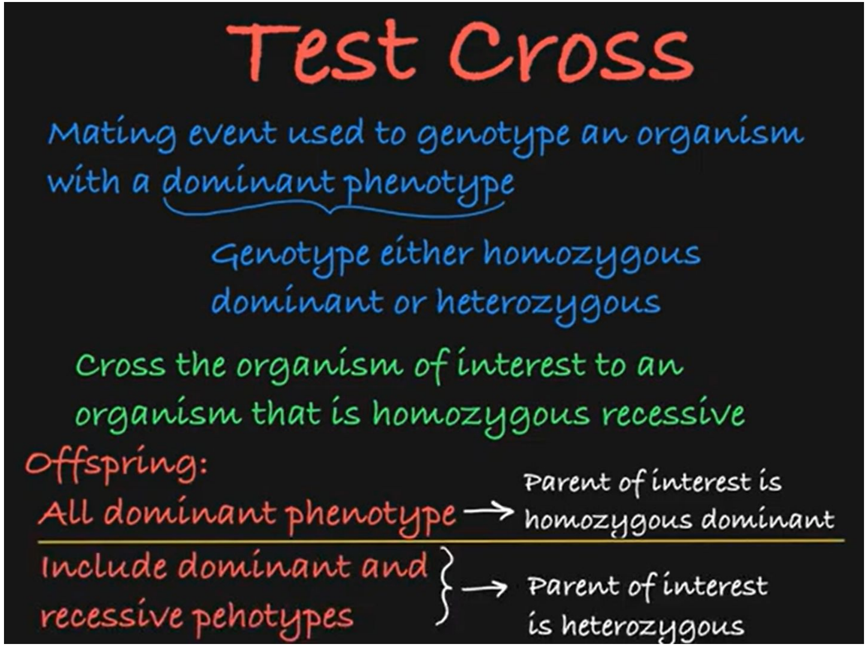
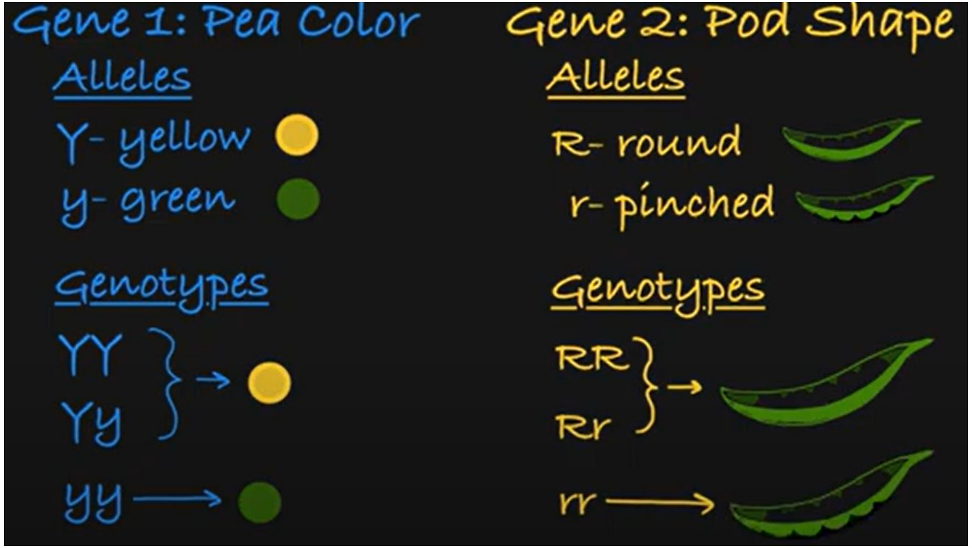




2. A typical dihybrid cross is conducted starting from pure breeding parents (AABB) and (aabb). All the F2 progeny that show the dominant AB phenotype are subjected to test cross. *__In what proportion of these test crosses__* you will find all the progeny to be of the AB phenotype? Explain. (3 marks)





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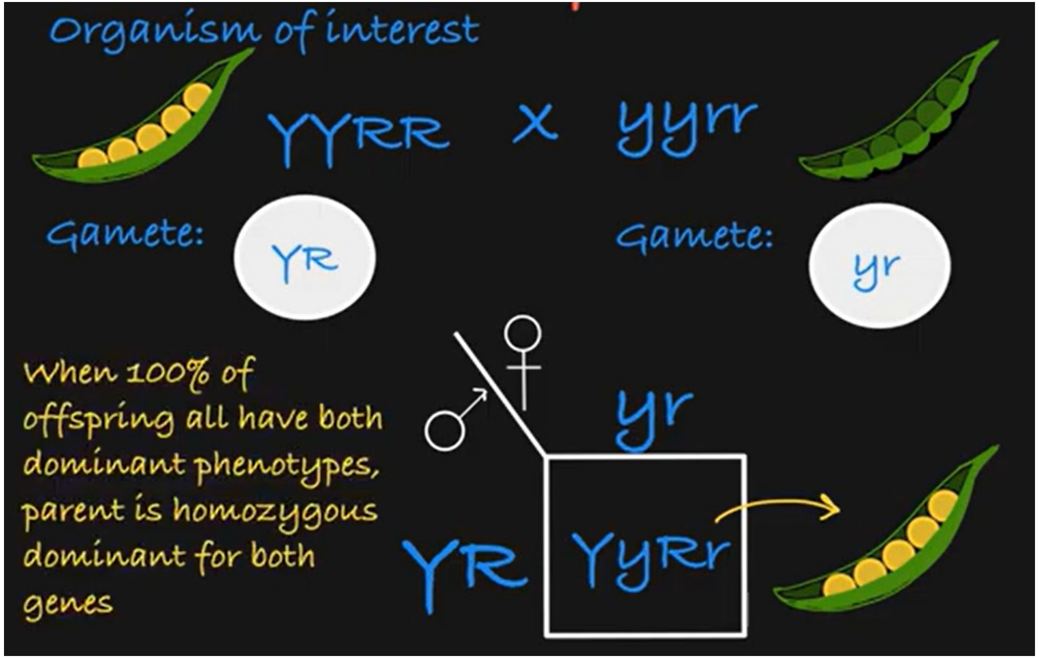

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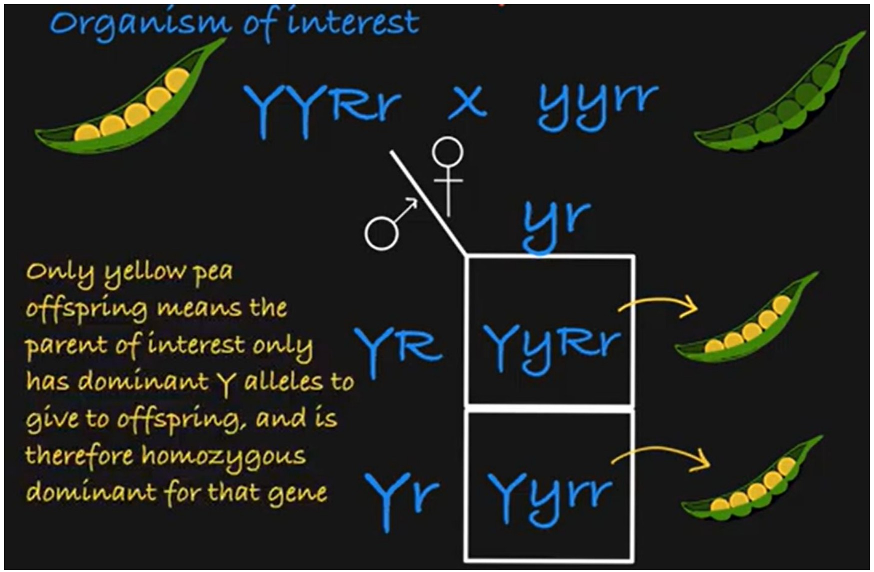

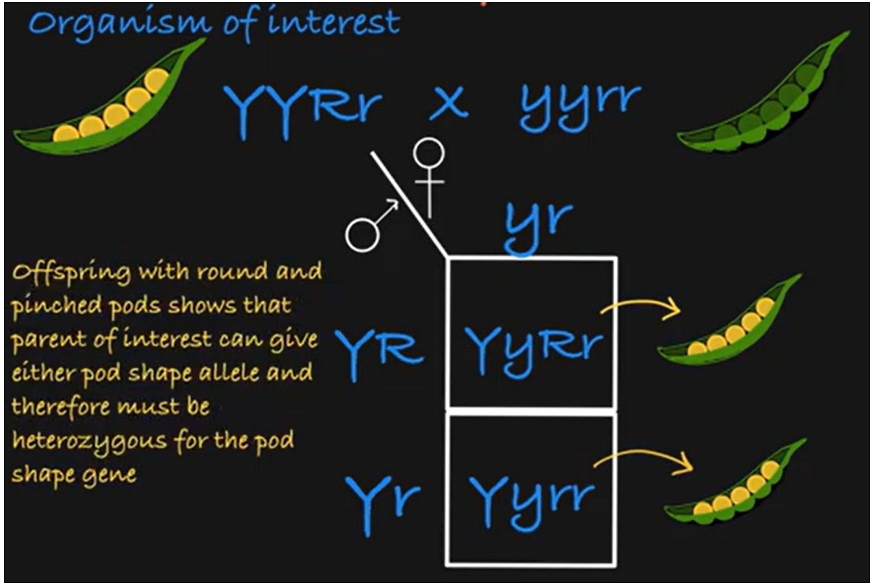

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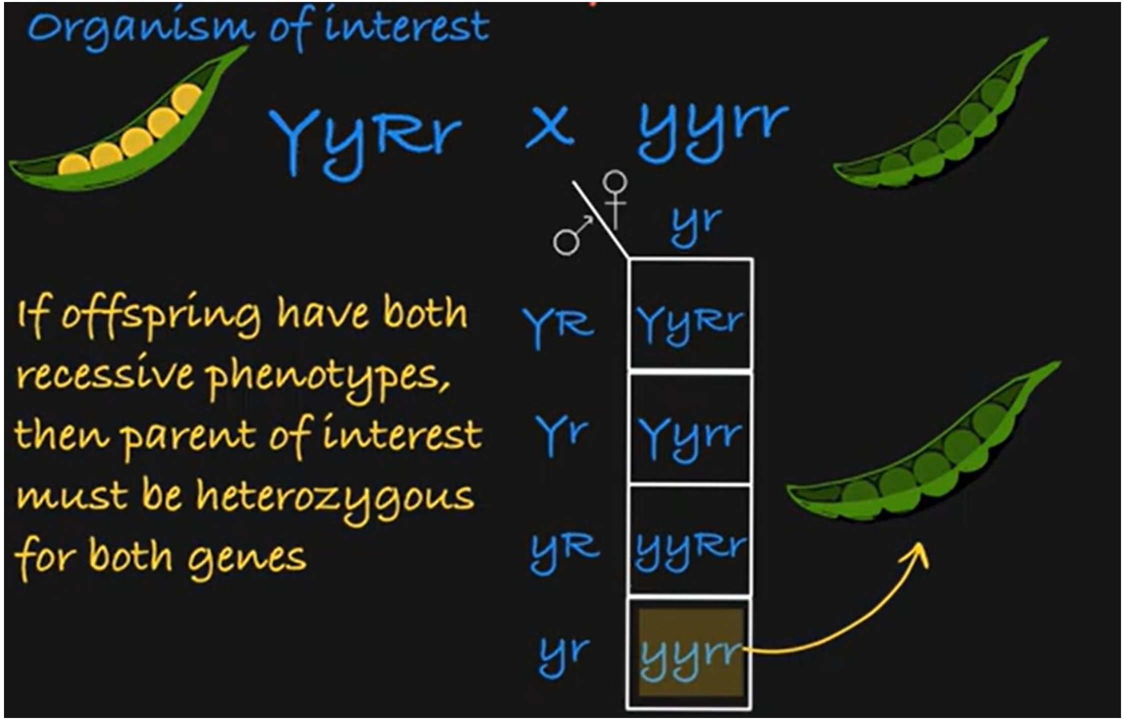


4. A woman with blood type O has a child with blood type O. She claims that a friend of hers is the child's father. In each of the following cases, explain your answer

   a. His blood type is A. Can he be excluded as the father on this evidence alone? (1 marks)

   b. Does the fact that the accused man's mother is type A and his father is type AB permit him to be excluded? (2 marks)

   c. Does the additional information that his mother's parents are both AB permit him to be excluded? (2 marks)

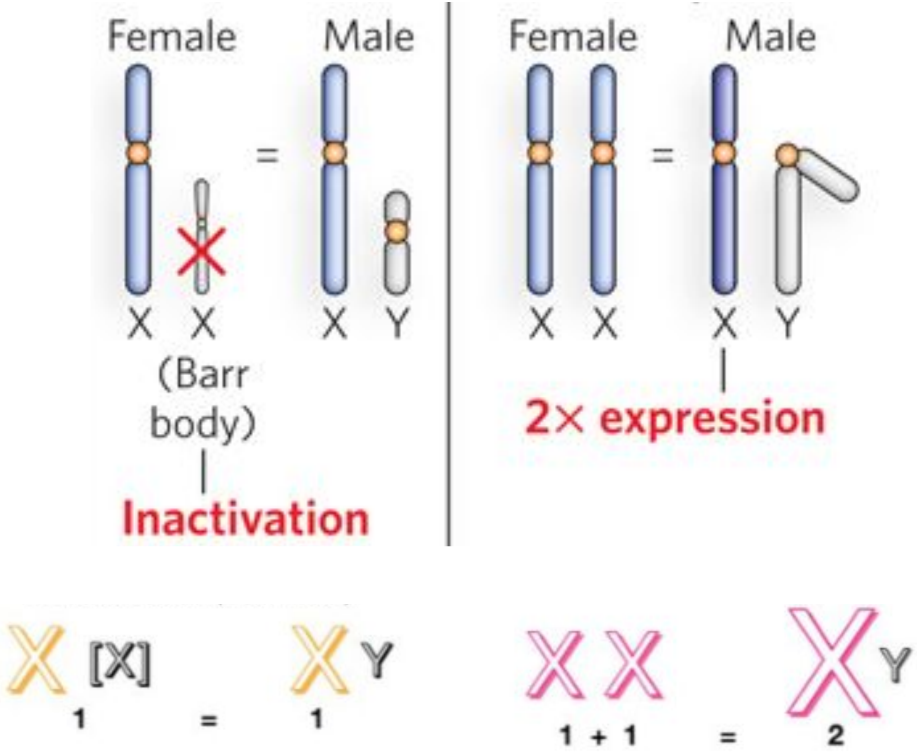
	I^A	I^B	i
I^A	$I^A I^A$ A	$I^A I^B$ AB	$I^A i$ A
I^B	$I^B I^A$ AB	$I^B I^B$ B	$I^B i$ B
i	$i\ I^A$ A	$i\ I^B$ B	$i\ i$ O

phenotype (blood type)	genotype
Type A	$I^A I^A$ $I^A i$
Type B	$I^B I^B$ $I^B i$
Type AB	$I^B I^A$
Type O	$i\ i$

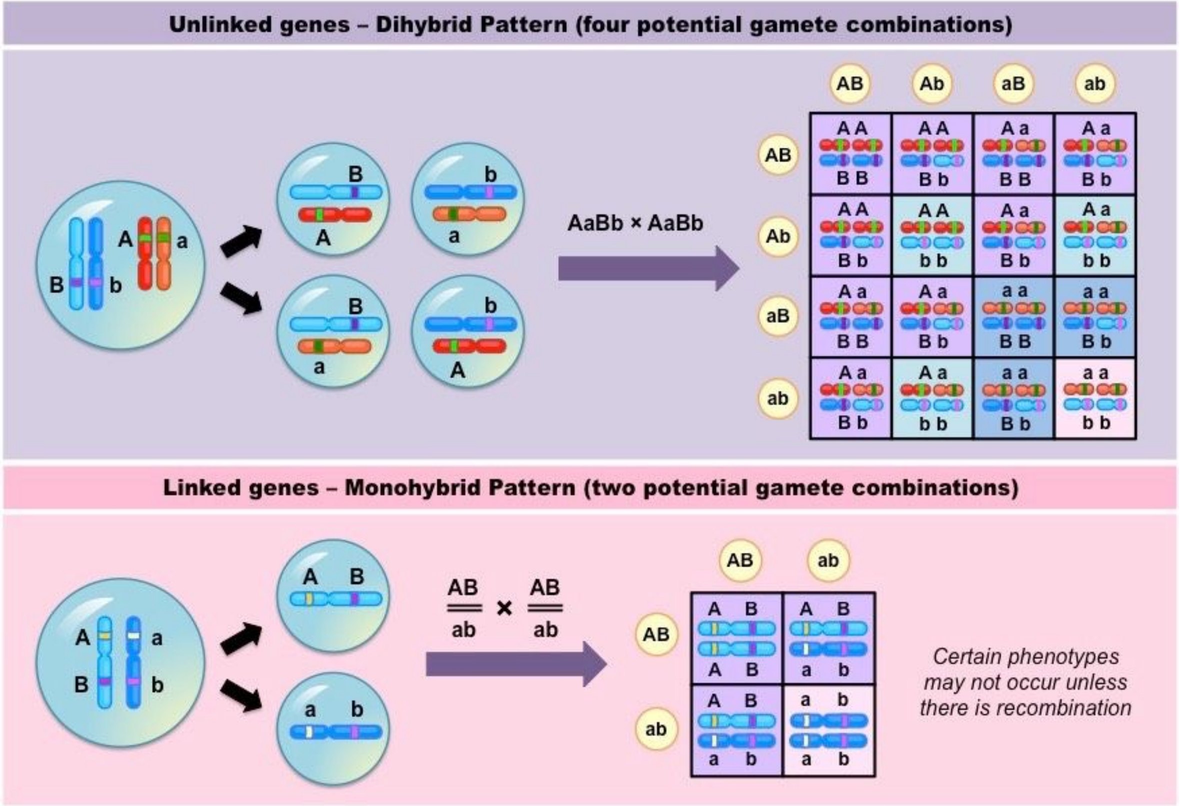
Blood Type Alleles:

I^A (co-dominant)
I^B (co-dominant)
i (recessive)

1. Define "Dosage Compensation". Explain the different ways in which dosage compensation occurs.



6. What is linkage? What causes linkage? Explain how you would detect linkage between any two loci.

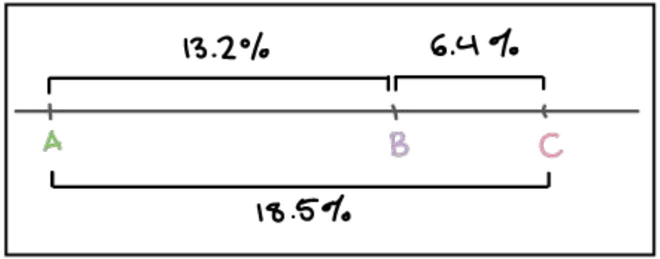


7. The genetic makeup of a population of humans has four different alleles for a certain X-linked gene and five different alleles for an autosomal locus. In relation to these two loci, what is the number of genetically different males in the population? (2 marks)

8. Crossover map distances determined by two-point crosses are P—C = 7, S—M = 10, C—M = 8, S—C = 2, and P—S = 5. Create a single linkage showing the relative position and distances between these loci. (2 marks)

RF (A - B) = 13.2 %
RF (B - C) = 6.4 %
RF (A - C) = 18.5 %

largest RF = outermost genes of trio



9. Trihybrid geranium plants with hairy leaves, blue flowers, and tall-growth habit are testcrossed with plants having smooth leaves, red flowers, and short-growth habit. The progeny are listed in the following table. Create a linkage map and calculate interference for these loci. (5 marks)

Leaf	**Flower**	**Height**	**Number**
Hairy	Blue	Tall	300
Hairy	Blue	Short	10
Hairy	Red	Tall	1,500
Hairy	Red	Short	180
Smooth	Blue	Tall	190
Smooth	Blue	Short	1,450
Smooth	Red	Tall	20
Smooth	Red	Short	350
Total			4,000

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Smooth	Red	Short	350
Total			4,000

Step 1: determine the Parental and Recombinant types

- Parentals
- Single Recombinants (Single Cross Overs)
- Double Recombinants (Double Cross Overs)

Step 2: Determine the Gene Order.

Step 3: Determine the linkage distances.

Step 4: Draw the map.

Step 5: Calculate Interference.

Interference (I) = 1- Coefficient of Coincidence

$$\text{C.o.C} = \frac{\text{Observed number of double recombinants}}{\text{Expected number of double recombinants}}$$

5. In poultry, R_P_ (where R and P are dominant alleles) gives rise to walnut comb. R_pp gives pea comb; rrP_ leads to rose comb and rrpp leads to single comb. You are provided a male with walnut comb (who you are told is heterozygous at both loci) and a pure breeding rose female. Write down the crosses you will do and the number of generations that it will take for you to get pure breeding pea comb individuals. Hint: You can cross the walnut comb male from the parental generation with the offspring generation as well. (4 marks)